Lecture: Old Trade Theory continues

A big **Thank You** to those of you who have actively participated in the lectures so far

Recap from last week

Recap from last week

- Three kinds of trade (similar goods, dissimilar goods, outsourcing by firms).

- Theory of absolute advantage: free trade is mutually beneficial as long as absolute production costs of goods differs across countries.

- Theory of comparative advantage: free trade is mutually beneficial as long as relative production cost of goods differs across countries.

Recap from last week

- Three kinds of trade (similar goods, dissimilar goods, outsourcing by firms).
- Theory of absolute advantage: free trade is mutually beneficial as long as absolute production costs of goods differs across countries.
- Theory of comparative advantage: free trade is mutually beneficial as long as relative production cost of goods differs across countries.
- Shortcoming I: Trade liberalisation has little social pain
- Shortcoming II: They don’t say much about about long-term term impact of trade.

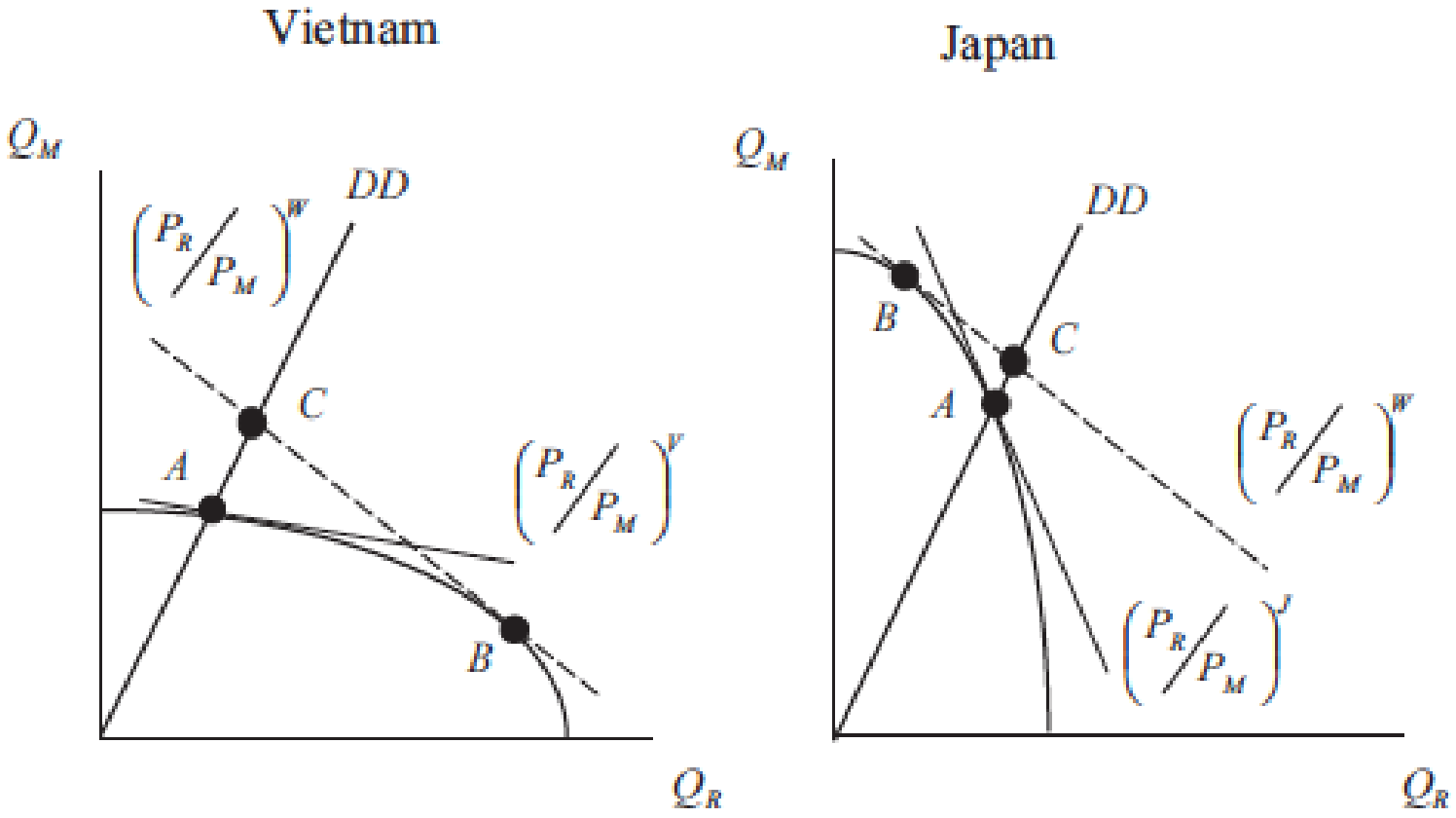


Figure 3.4. Autarky and Comparative Advantage in Vietnam and Japan

Recap from last week

- Three kinds of trade (similar goods, dissimilar goods, outsourcing by firms).

- Theory of absolute advantage: free trade is mutually beneficial as long as absolute production costs of goods differs across countries.

- Theory of comparative advantage: free trade is mu[illegible]eficial as long as relative production cost of goods differs across countries.

- Shortcoming I: Trad[illegible]ation has little social pain

- Shortcoming II: They don't say much about about long-term term impact of trade.

So we need a more advanced theory!

Lecture plan:

- Modern theory of comparative advantage (H-O model).
- Winners and losers of free trade inside the countries according to HO.
- Critiques of modern theory of comparative advantage.

Lecture plan:

- **Modern theory of comparative advantage (H-O model).**
- Winners and losers of free trade inside the countries according to HO.
- Critiques of modern theory of comparative advantage.

Heckscher-Ohlin Model:

Difference in production costs
(**Relative cost of labor, capital, land** & production knowledge)

Absolute/Relative Price difference

Differences in demand

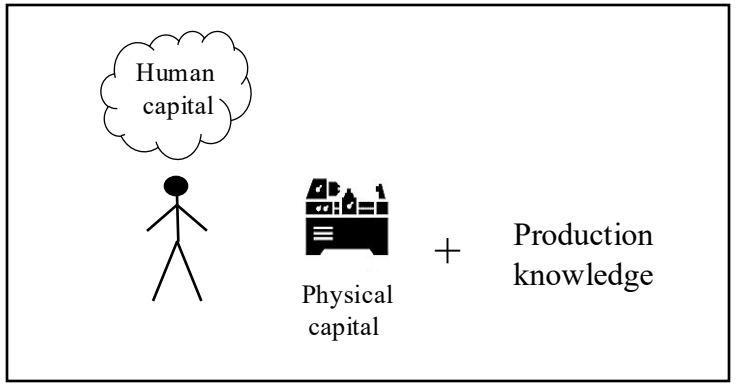
Heckscher-Ohlin Model:

Difference in production costs
(**Relative cost of labor, capital, land** & production knowledge)

Absolute/Relative Price difference

Differences in demand

Firms



Land & natural resources

Heckscher Ohin model (modern version of comparative advantage):

Which sectors do developing and rich countries specialize in when they open up?

Heckscher Ohin model (modern version of comparative advantage):

Which sectors do developing and rich countries specialize in when they open up?

Knowledge-intensive industries

Capital-intensive industries

Labor-intensive industries

Agriculture &commodities

Agriculture & commodities

Coffee (commodities)

Agricultural goods

Labor-intensive industries (light industries)

Textile

Assembly of electronic goods

Capital-intensive industries (heavy industries)

Steel

Production of car engines

knowledge-intensive industries

Computer chip production

Product design

Heckscher Ohin model (modern version of comparative advantage):

Which sectors do developing and **rich countries** specialize in when they open up?

Knowledge-intensive industries

Capital-intensive industries

Labor-intensive industries

Agriculture &commodities

Heckscher Ohin model (modern version of comparative advantage):

Which sectors do developing and **rich countries** specialize in when they open up?

$$\text{Relative factor endowments} = \frac{\text{Land stock}, \overset{\textit{High}}{\text{Physical capital stock}}, \overset{\textit{High}}{\text{Human capital stock}}}{\text{Labor stock}} \Rightarrow \text{Relative input costs}$$

Heckscher Ohin model (modern version of comparative advantage):

Which sectors do developing and **rich countries** specialize in when they open up?

$$\text{Relative factor endowments} = \frac{\text{Land stock}, \overset{\textit{High}}{\text{Physical capital stock}}, \overset{\textit{High}}{\text{Human capital stock}}}{\text{Labor stock}} \Rightarrow \text{Relative input costs}$$

Exports

Knowledge-intensive industries

Capital-intensive industries

Imports

Labor-intensive industries

Agriculture &commodities

Heckscher Ohin model (modern version of comparative advantage):

Which sectors do **developing** and rich countries specialize in when they open up?

$$\text{Relative factor endowments} = \frac{\text{Land stock}, \overset{\textit{Low}}{\text{Physical capital stock}}, \overset{\textit{Low}}{\text{Human capital stock}}}{\text{Labor stock}} \Rightarrow \text{Relative input costs}$$

Imports

Knowledge-intensive industries

Capital-intensive industries

Exports

Labor-intensive industries

Agriculture &commodities

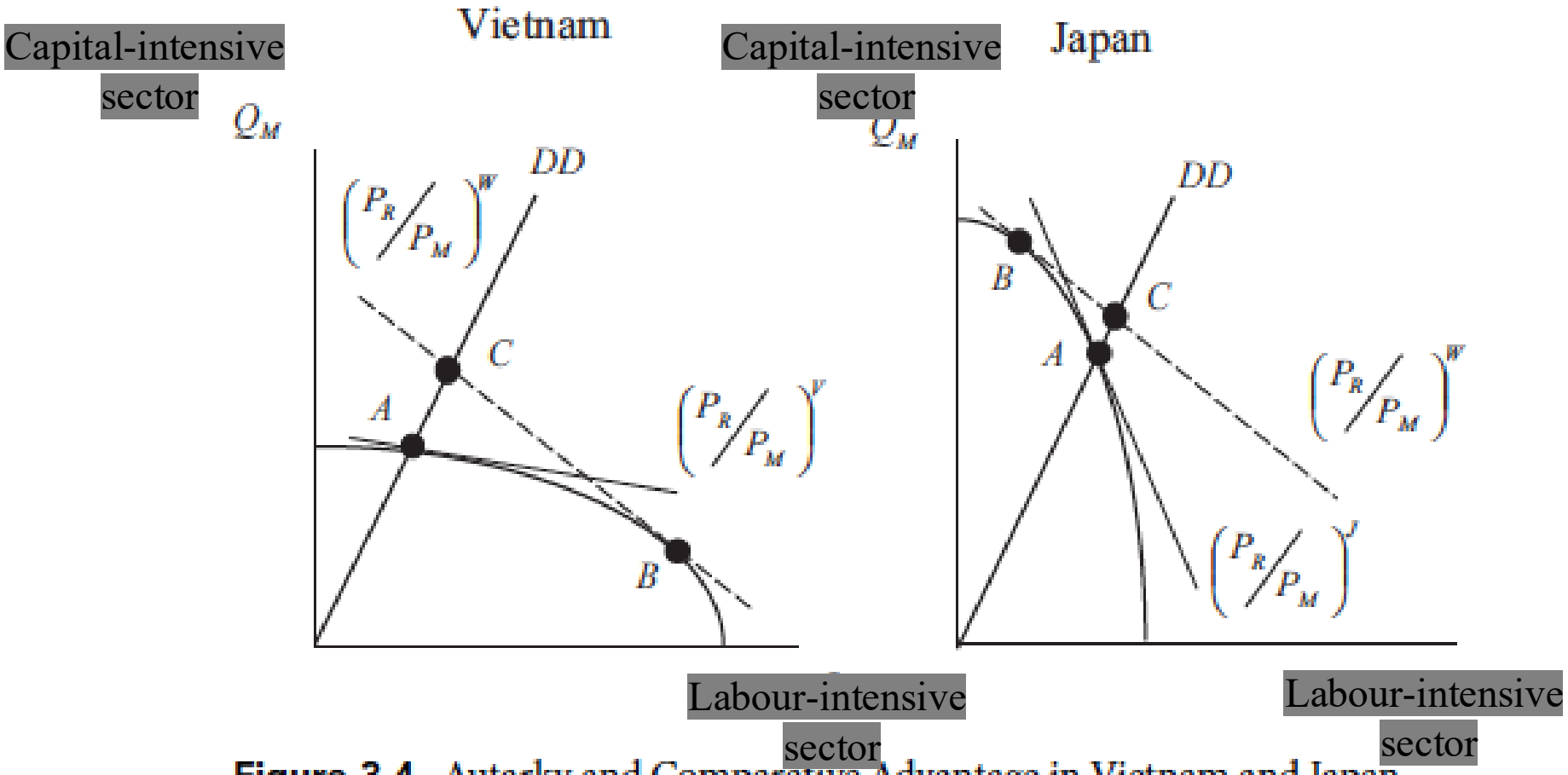


Figure 3.4. Autarky and Comparative Advantage in Vietnam and Japan

Assumptions of the HO's model:

Relative difference in production costs

?

(**Input costs**, Technology)

→

Relative Price difference

←

Differences in relative demand

Assumptions of the HO's model:

Relative difference in production costs

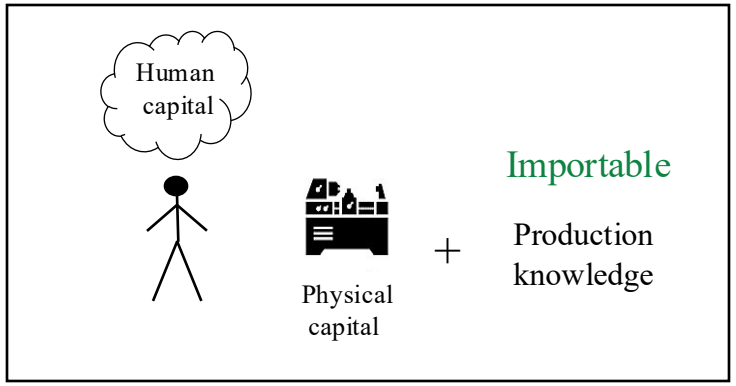
Importable

(**Input costs**, Technology)

Relative Price difference

Differences in relative demand

Firms



Land & natural resources

Assumptions of the HO's model:

Relative difference in production costs → *Relative* Price difference ← Differences in relative demand

Importable

(**Input costs**, Technology)

- Technology/knowledge of production is importable. Production capability changes overtime (Dynamic)

Assumptions of the HO's model:

Relative difference in production costs → *Relative* Price difference ← Differences in relative demand

Same

(**Input costs**, Technology)

- Technology/knowledge of production is importable. Production capability changes overtime (Dynamic)
- Taste is the same in two countries (composition of their consumption basket is the same).
- Perfect competition (so constant return to scale)
- No transportation cost
- No transaction cost
- No government interference in prices

Assumptions of the HO's model:

Relative difference in production costs → *Relative* Price difference ← Differences in relative demand

Same

(**Input costs**, Technology)

- Technology/knowledge of production is importable. Production capability changes overtime (Dynamic)
- Taste is the same in two countries (composition of their consumption basket is the same).
- Perfect competition (so constant return to scale)
- No transportation cost
- No transaction cost
- No government interference in prices
- Factor mobility inside the country (but not between countries).

Assumptions of the HO's model:

Relative difference in production costs → *Relative* Price difference ← Differences in relative demand

Same

(**Input costs**, Technology)

- Technology/knowledge of production is importable. Production capability changes overtime (Dynamic)
- Taste is the same in two countries (composition of their consumption basket is the same).
- Perfect competition (so constant return to scale)
- No transportation cost
- No transaction cost
- No government interference in prices
- Factor mobility inside the country (but not between countries).

Why?

China's Huajian Group boost Ethiopian manufacturing sector

CGTN

Subscribe 112,334

9,897 views

Download Add to Share More 34 2

CLICK HERE

[redacted]

China's Huajian Group boost Ethiopian manufacturing sector

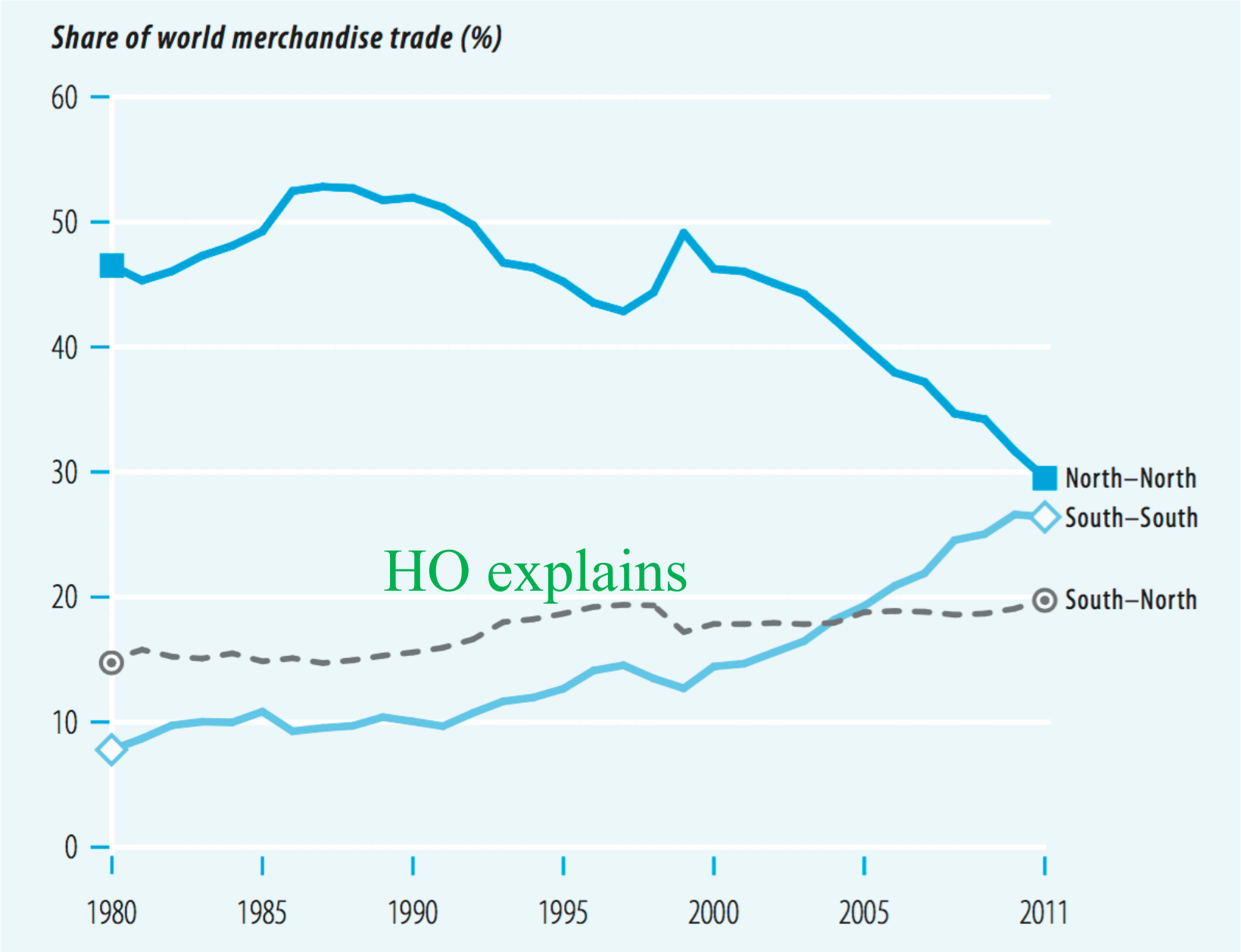
CGTN

Subscribe 112,334

9,897 views

Download Add to Share More 34 2

CLICK HERE



Note: North in 1980 refers to Australia, Canada, Japan, New Zealand, the United States and Western Europe.
Source: HDRO calculations based on UNSD (2012).

Table 4.1. Types of trade

	Type of trade	Phrase	Meaning	
HO explains	Inter-industry	Either/or	*Either* imports *or* exports in a given sector of the economy	North-South
	Horizontal intra-industry	Both/and/same	*Both* imports *and* exports in a given sector of the economy and at the *same* stage of processing	North-North
HO explains	Vertical intra-industry (Outsourcing by firms)	Both/and/different	*Both* imports *and* exports in a given sector of the economy and at *different* stages of processing	North-South

Lecture plan:

- Modern theory of comparative advantage (H-O model).
- **Winners and losers of free trade inside the countries according to HO.**
- Critiques of modern theory of comparative advantage.

Q: Who will be the winners and losers of free trade in rich and developing counties according to HO model?

Join at menti.com | use code 6895 9886

Who are among the winners of trade in rich countries according to the HO model?

Software engineers in Google	Car assembler in Detroit	Bank depositors in Germany	CEO of Tesco in London	Unskilled labor	Owners of physical capital
0	0	0	0	0	0

Responding is off

Press C to allow new responses

Press ENTER to show correct answer

Q: Who will be the winners and losers of free trade in rich and developing counties according to HO model?

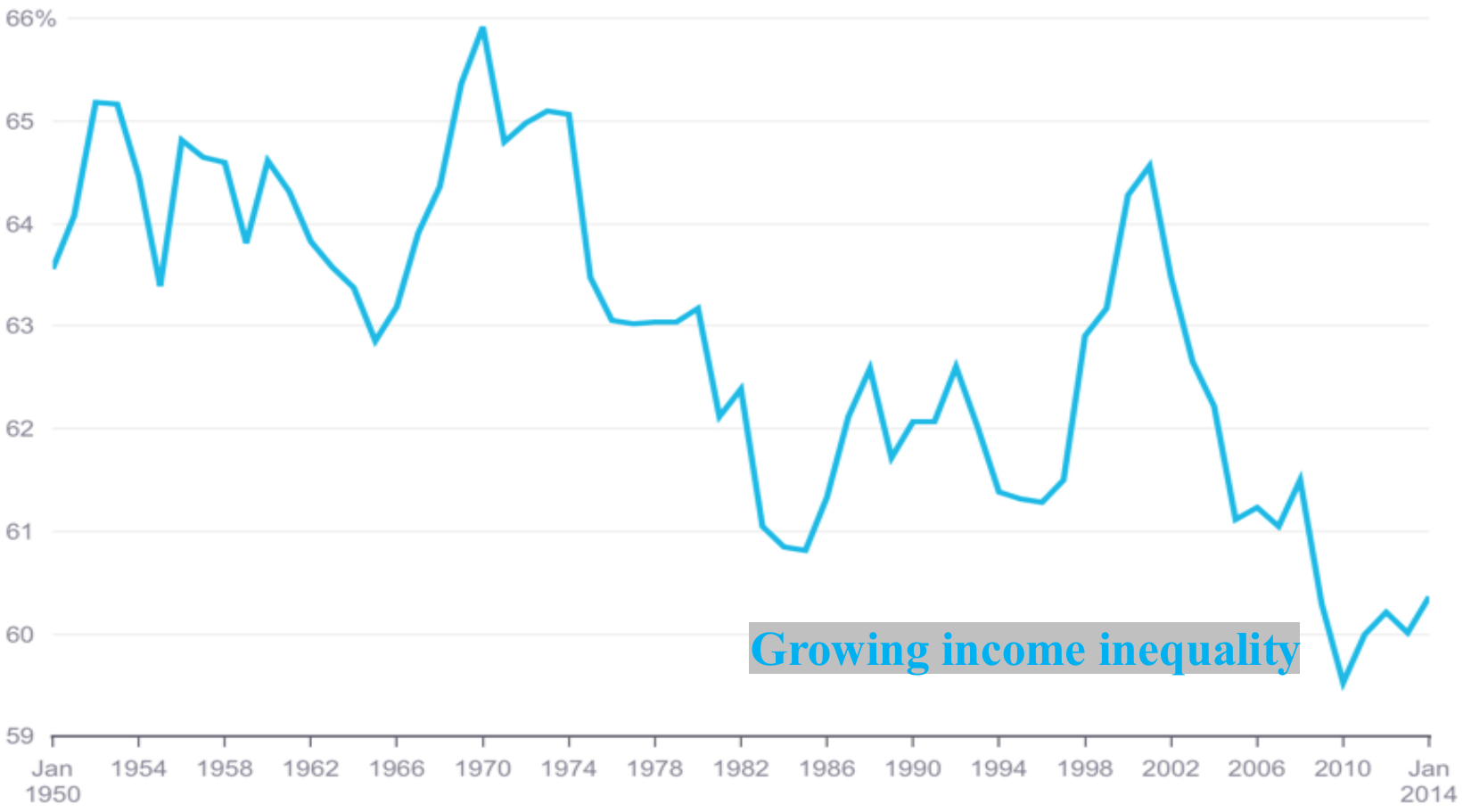
Q: Who will be the winners and losers of free trade in rich and developing counties according to HO model?

Developed counties: Relative price for capital-intensive & knowledge intensive- good increases. Hence, this capital-intensive sectors expands and labor-intensive sectors decline. Demand for human & physical capital goes up while demand for unskilled workers go down. Resource reallocation across sectors.

Eminem in a car factory in Detroit

Less of the Pie for Labor

Share of GDP received by workers. **in the US**



Source: Federal Reserve Bank of St. Louis

Less of the Pie for Labor

Share of GDP received by workers. **in the US**

Why does this indicate growing income inequality?

Source: Federal Reserve Bank of St. Louis

Q: Who will be the winners and losers of free trade in rich and developing counties according to HO model?

Developed counties: Relative price for capital-intensive good increases. Hence, this capital-intensive sectors expands and labor-intensive sectors decline. Demand for human & physical capital goes up while demand for unskilled workers go down. Resource reallocation across sectors.

Developing counties: Relative price for labor-intensive good increases. Hence, this labor-intensive sectors expands and capital-intensive sectors decline. Demand for unskilled labor goes up while demand for human & physical capital goes down. Resource reallocation across sectors.

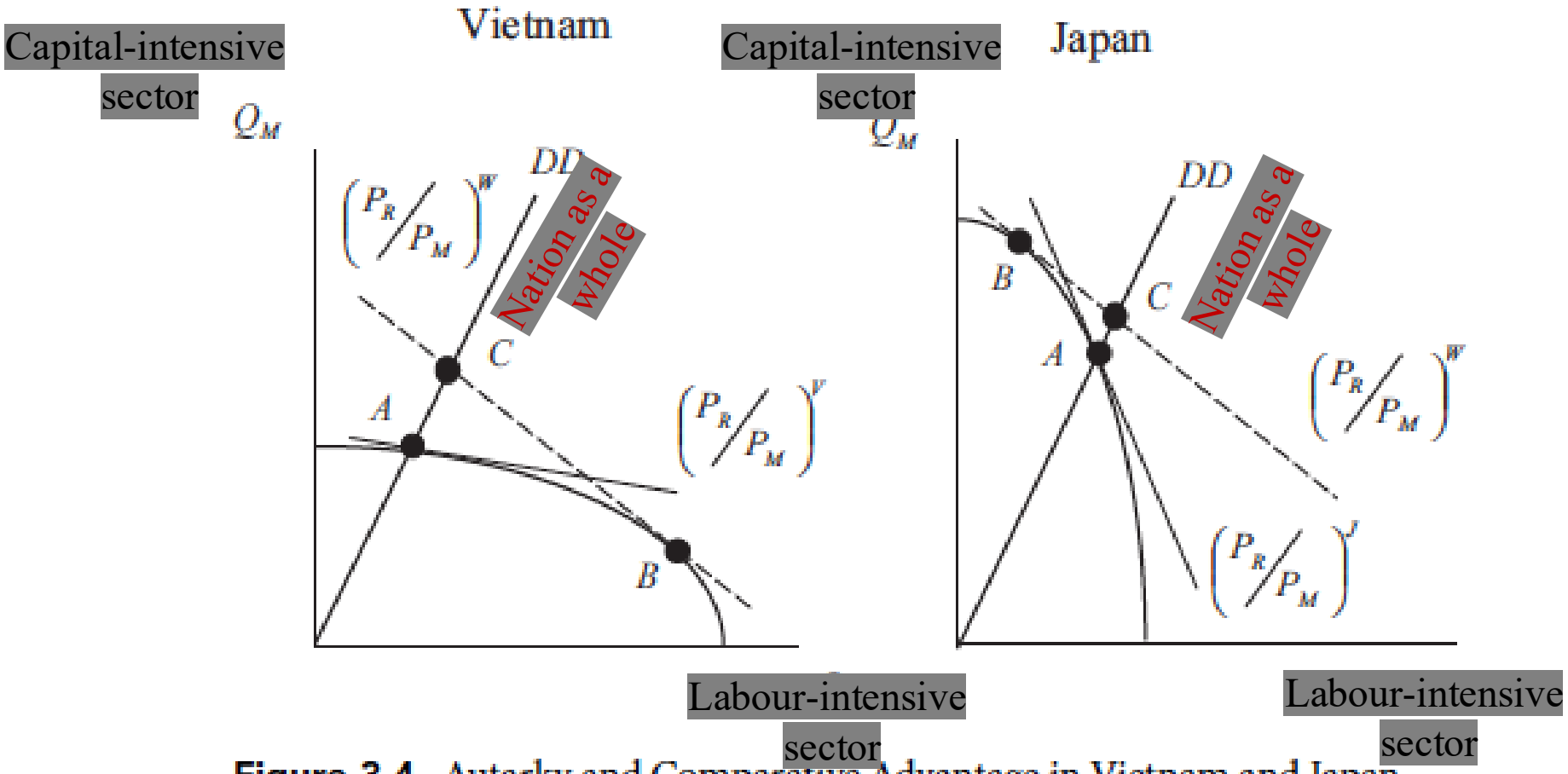


Figure 3.4. Autarky and Comparative Advantage in Vietnam and Japan

Q: Who will be the winners and losers of free trade in rich and developing counties according to HO model?

Developed counties: Relative price for capital-intensive good increases. Hence, this capital-intensive sectors expands and labor-intensive sectors decline. Demand for human & physical capital goes up while demand for unskilled workers go down. Resource reallocation across sectors.

Developing counties: Relative price for labor-intensive good increases. Hence, this labor-intensive sectors expands and capital-intensive sectors decline. Demand for unskilled labor goes up while demand for human & physical capital goes down. Resource reallocation across sectors.

If winners compensate the losers, then we will have a Pareto improving outcome (increase in net income).

So trade can *potentially* make everyone better-off.

Q: Who will be the winners and losers of free trade in rich and developing counties according to HO model?

Developed counties: Relative price for capital-intensive good increases. Hence, this capital-intensive sectors expands and labor-intensive sectors decline. Demand for human & physical capital goes up while demand for unskilled workers go down. Resource reallocation across sectors.

Developing counties: Relative price for labor-intensive good increases. Hence, this labor-intensive sectors expands and capital-intensive sectors decline. Demand for unskilled labor goes up while demand for human & physical capital goes down. Resource reallocation across sectors.

If winners compensate the losers, then we will have a Pareto improving outcome (increase in net income).

So trade can *potentially* make everyone better-off.

You Retweeted

WTO @wto · Sep 26

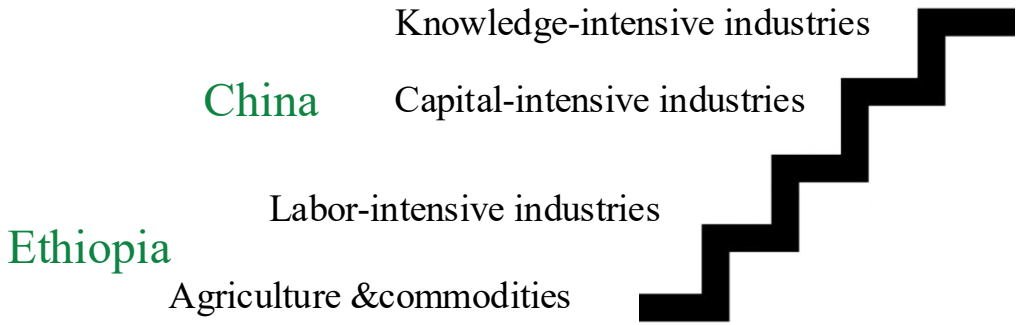
What economics textbooks got right about global trade and what politics got wrong. @paulkrugman tells it all at the #WTOPublicForum

Join at menti.com | use code 6895 9886

What will happen to Ethiopia's shoe production in the long-term? Explain.

Long-term predictions of the model?

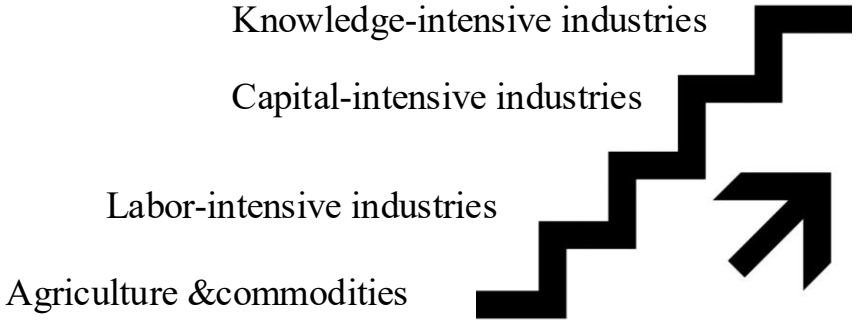
Long-term predictions of the model?



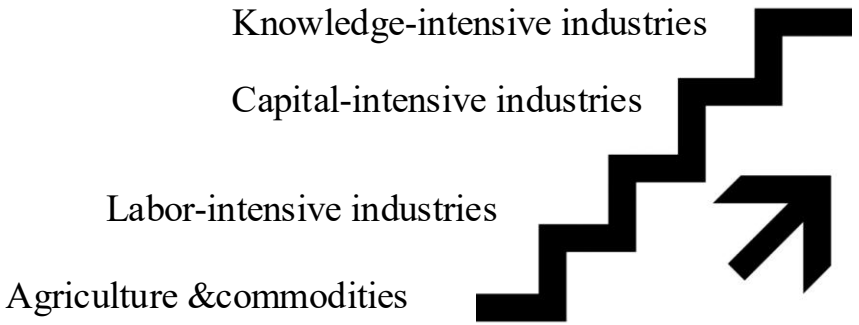
The Economist
Naval gazing—Russia eyeballs Ukraine
Edit genes, but not like this
Technology Quarterly: Zero carbon
Our books of the year
Chip wars
America, China and silicon supremacy

The Economist
The world's least successful president
Putin threatens Belarus
Pakistan: impoverished by its army
How the mighty dollar falls
JANUARY 12TH–18TH 2019
Red moon rising
Will China dominate science?

Long-term predictions of the model



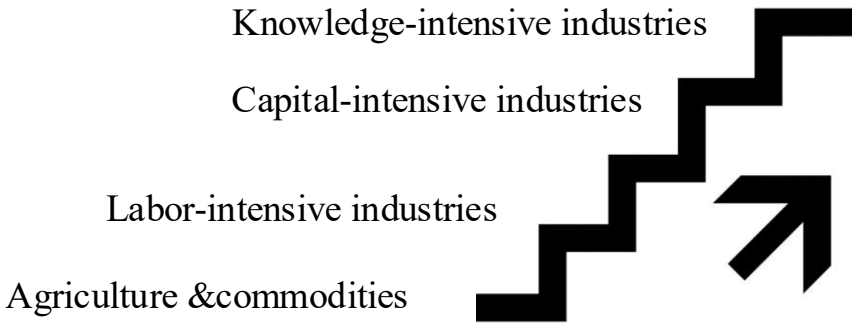
Long-term predictions of the model



Policy implication: **open up your economy to free trade & multinationals**

Long-term predictions of the model

Advantage of backwardness
(favorable inputs+ access to latest knowledge production)
Going-up the global value chain
Automatic catch-up in income (convergence)



Policy implication: **open up your economy to free trade & multinationals**

Lecture plan:

- Modern theory of comparative advantage (H-O model).
- Winners and losers of free trade inside the countries according to HO.
- **Critiques of modern theory of comparative advantage.**

Shortcoming I: You can’t import/learn the technology right away

Assumptions of the HO's model:

Relative difference in production costs → *Relative* Price difference ← Differences in relative demand

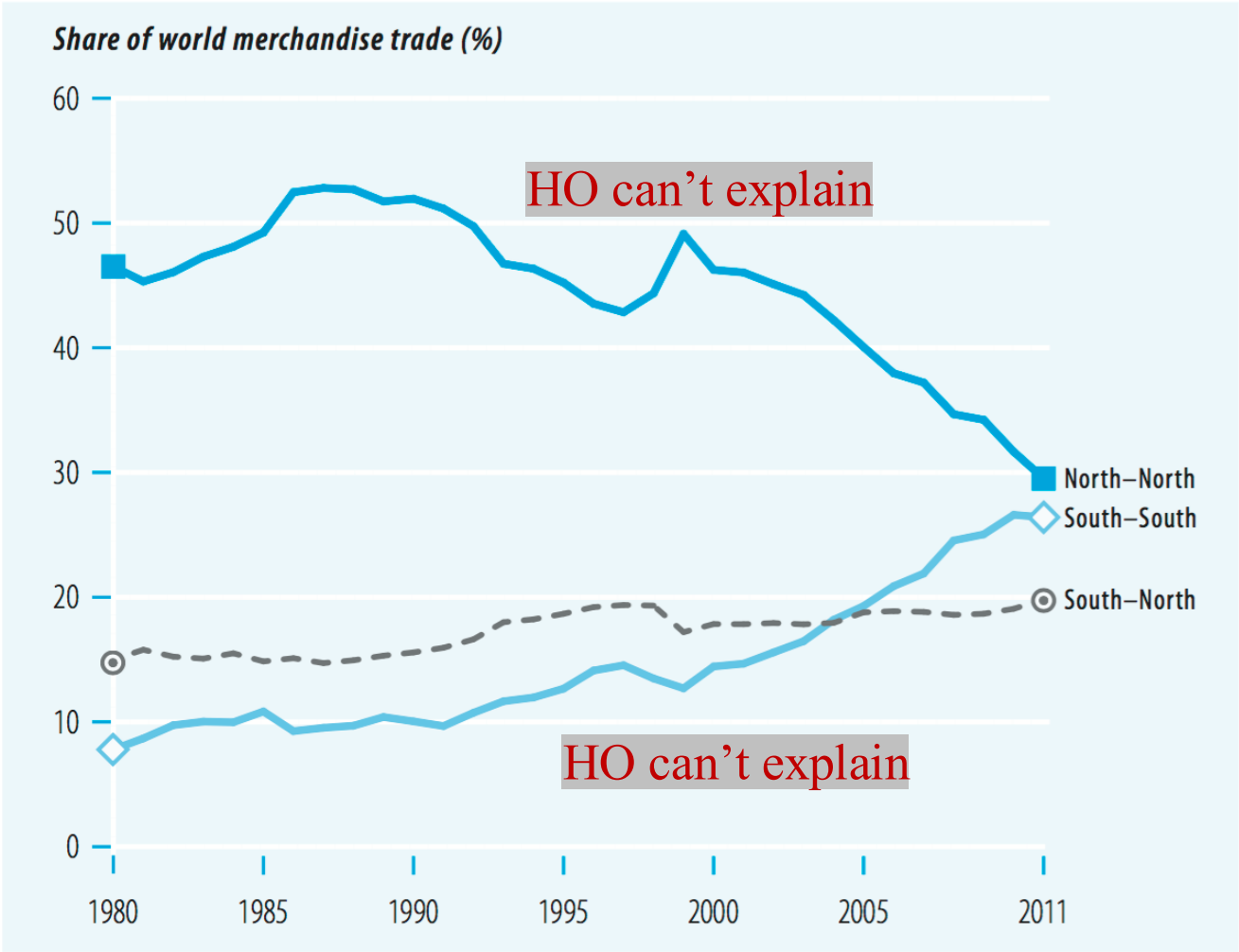
Same

(Input costs, Technology)

Really?!

- Technology/knowledge of production is importable.

Shortcoming II:



Note: North in 1980 refers to Australia, Canada, Japan, New Zealand, the United States and Western Europe.
Source: HDRO calculations based on UNSD (2012).

Table 4.1. Types of trade

	Type of trade	Phrase	Meaning	
HO explains	Inter-industry	Either/or	*Either* imports *or* exports in a given sector of the economy	North-South
HO can't explain	Horizontal intra-industry	Both/and/same	*Both* imports *and* exports in a given sector of the economy and at the *same* stage of processing	North-North
HO explains	Vertical intra-industry (Outsourcing by firms)	Both/and/different	*Both* imports *and* exports in a given sector of the economy and at *different* stages of processing	North-South

Summary

Summary

- HO explains North-South trade & trade in *dissimilar* goods and tasks across countries.

Summary

- HO explains North-South trade & trade in *dissimilar* goods and tasks across countries.

- HO partially explains the winners & losers of globalisation. But everyone *could potentially* become better-off by free trade, if winners had compensated the losers.

Summary

- HO explains North-South trade & trade in *dissimilar* goods and tasks across countries.
- HO partially explains the winners & losers of globalisation. But everyone *could potentially* become better-off by free trade, if winners had compensated the losers.
- Policy implication of HO: open up your economy to trade and multinationals. Convergence in income in the long-term.

Summary

- HO explains North-South trade & trade in *dissimilar* goods and tasks across countries.
- HO partially explains the winners & losers of globalisation. But everyone *could potentially* become better-off by free trade, if winners had compensated the losers.
- Policy implication of HO: open up your economy to trade and multinationals. Convergence in income in the long-term.
- HO can’t explain North-North trade and trade in similar goods and tasks across countries.

Summary

- HO explains North-South trade & trade in *dissimilar* goods and tasks across countries.
- HO partially explains the winners & losers of globalisation. But everyone *could potentially* become better-off by free trade, if winners had compensated the losers.
- Policy implication of HO: open up your economy to trade and multinationals. Convergence in income in the long-term.
- HO can't explain North-North trade and trade in similar goods and tasks across countries.
- HO ignores the problems associated with technological learning.

Appendix

Stolper–Samuelson theorem

From Wikipedia, the free encyclopedia

The **Stolper–Samuelson theorem** is a theorem in Heckscher–Ohlin trade theory. It describes the relationship between relative prices of output and relative factor returns—specifically, real wages and real returns to capital.

The theorem states that—under specific economic assumptions (constant returns to scale, perfect competition, equality of the number of factors to the number of products)—a rise in the relative price of a good will lead to a rise in the real return to that factor which is used most intensively in the production of the good, and conversely, to a fall in the real return to the other factor.

Derivation [edit]

Considering a two-good economy that produces only wheat and cloth, with labor and land being the only factors of production, wheat a land-intensive industry and cloth a labor-intensive one, and assuming that the price of each product equals its marginal cost, the theorem can be derived.

The price of cloth should be:

(1) $P(C) = ar + bw,$

with *P*(*C*) standing for the price of cloth, *r* standing for rent paid to landowners, *w* for wage levels and *a* and *b* respectively standing for the amount of land and labor used, and do not change with the prices of goods.

Similarly, the price of wheat would be:

(2) $P(W) = cr + dw$

with *P*(*W*) standing for the price of wheat, *r* and *w* for rent and wages, and *c* and *d* for the respective amount of land and labor used, and also considered to be constant.

If, then, cloth experiences a rise in its price, at least one of its factors must also become more expensive, for equation 1 to hold true, since the relative amounts of labor and land are not affected by changing prices. It can be assumed that it would be labor—the factor that is intensively used in the production of cloth—that would rise.

When wages rise, rent must fall, in order for equation 2 to hold true. But a fall in rent also affects equation 1. For it to still hold true, then, the rise in wages must be more than proportional to the rise in cloth prices.

A rise in the price of a product, then, will more than proportionally raise the return to the most intensively used factor, and decrease the return to the less intensively used factor.